

Problems on Basic Probability Theory

Problem Set 1: Alarm System in a Neighborhood

A security company monitors houses for burglaries using an alarm system. The probabilities are given in the joint probability table below:

	Alarm	$\neg$ Alarm
Burglary	0.09	0.01
$\neg$ Burglary	0.10	0.80

- (a) Compute the marginal probability $P(\text{alarm})$.
- (b) Compute the conditional probability $P(\text{burglary} \mid \text{alarm})$.
- (c) Find the joint probability $P(\text{burglary} \wedge \neg \text{alarm})$.

Problem Set 2: Applying Probability Rules

Using the same alarm-burglary data:

- (a) Use the product rule to compute $P(\text{alarm} \wedge \text{burglary})$.
- (b) Find $P(\text{alarm} \mid \text{burglary})$.
- (c) Verify that $P(\text{alarm} \wedge \text{burglary}) = P(\text{alarm} \mid \text{burglary}) \times P(\text{burglary})$.

Problem Set 3: Students and Exam Preparedness

Given the following joint distribution for attributes of students:

	Smart & Study	Smart & $\neg$ Study	$\neg$ Smart & Study	$\neg$ Smart & $\neg$ Study
Prepared	0.432	0.16	0.084	0.008
$\neg$ Prepared	0.048	0.16	0.036	0.072

- (a) What is the prior probability $P(\text{smart})$?
- (b) What is the prior probability $P(\text{study})$?
- (c) What is the conditional probability $P(\text{prepared} \mid \text{smart}, \text{study})$?

Problem Set 4: Exploring Independence

Based on the same table above:

- (a) Are “smart” and “study” independent? Justify using the definition of independence.
- (b) Are “prepared” and “study” independent? Show your calculations.
- (c) Discuss whether assuming independence could lead to faulty inference in this context.

Problem Set 5: Conditional Independence

- (a) Is “smart” conditionally independent of “prepared” given “study”? Justify mathematically.
- (b) Is “study” conditionally independent of “prepared” given “smart”?
- (c) How does assuming conditional independence help in simplifying joint distribution computation?

Problem Set 6: Moon Phase, Burglary, and Light-Level

In a separate model, Moon-Phase and Burglary are conditionally independent given Light-Level. That is,

$$P(\text{Moon} \wedge \text{Burglary} \mid \text{Light}) = P(\text{Moon} \mid \text{Light}) \cdot P(\text{Burglary} \mid \text{Light})$$

- (a) Explain the meaning of conditional independence in this scenario.
- (b) If $P(\text{Moon} \mid \text{Light}) = 0.3$ and $P(\text{Burglary} \mid \text{Light}) = 0.2$, compute $P(\text{Moon} \wedge \text{Burglary} \mid \text{Light})$.
- (c) How would this decomposition help in computing the full joint distribution $P(\text{Moon}, \text{Burglary}, \text{Light})$?